

## 4.1 Introduction

Performing a graft is only half the process—the **real success depends on proper aftercare and monitoring**. Even a perfectly executed graft can fail if it is not properly maintained during the healing phase.

This module focuses on:

- Post-grafting care
  - Identifying and fixing problems
  - Advanced practices such as **topworking** and **interstocks**
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## 4.2 The Healing Process After Grafting

After the scion and rootstock are joined, the plant begins a natural healing process:

### Stages of Healing

1. **Initial Contact**
    - Cambium layers are pressed together
    - Moisture must be retained
  2. **Callus Formation**
    - Soft tissue develops at the graft union
    - Bridges the gap between scion and rootstock
  3. **Vascular Connection**
    - Xylem and phloem reconnect
    - Water and nutrients begin flowing
  4. **New Growth**
    - Buds on the scion begin to grow
    - Indicates successful graft
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## 4.3 Immediate Post-Grafting Care

### 1. Securing the Graft

- Ensure wrapping is tight and stable
- Prevent movement that can break cambium contact

### 2. Moisture Control

- Prevent the graft from drying out

- Use wax or sealant to protect exposed areas

### 3. Environmental Protection

- Keep grafts in:
    - Mild temperatures
    - Partial shade
    - Low wind exposure
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## 4.4 Managing Growth After Grafting

### 1. Removing Suckers

Suckers are shoots that grow from the rootstock.

#### Why remove them?

- They compete for nutrients
  - Reduce energy available to the scion
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### 2. Supporting the Scion

- New shoots are fragile
  - Use stakes or supports if necessary
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### 3. Gradual Exposure

- Slowly introduce the plant to full sunlight
  - Prevent stress or dehydration
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## 4.5 Adjusting Wrapping Materials

As the graft heals:

- The union will **expand and thicken**
- Tight wraps can **girdle** (choke) the plant

#### Best Practice:

- Loosen or remove wraps after healing begins
  - Ensure they do not restrict growth
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## 4.6 Identifying Successful Grafts

A successful graft will show:

- Healthy, green scion
  - Bud growth and leaf development
  - Strong and stable union
  - Gradual thickening at the graft site
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## 4.7 Identifying Graft Failure

Recognizing failure early allows corrective action.

### Common Signs:

- Scion turns brown or black
  - No bud growth
  - Weak or detached union
  - Rootstock produces excessive shoots
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## 4.8 Common Causes of Failure

Based on grafting principles, failures usually result from:

### 1. Poor Cambium Alignment

- Prevents vascular connection

### 2. Drying of the Scion

- Occurs when sealing is inadequate

### 3. Disease Transmission

- Using infected plant material or dirty tools

### 4. Incompatibility

- Scion and rootstock are not genetically compatible

### 5. Environmental Stress

- Extreme heat, cold, or wind
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## 4.9 Troubleshooting Guide

| Problem                 | Likely Cause                     | Solution                      |
|-------------------------|----------------------------------|-------------------------------|
| Scion dries out         | Poor sealing or delay in joining | Seal immediately, work faster |
| No growth after weeks   | Cambium misalignment             | Improve cutting and alignment |
| Weak union              | Loose wrapping                   | Secure firmly                 |
| Excess rootstock shoots | Lack of pruning                  | Remove suckers regularly      |
| Infection signs         | Dirty tools                      | Sterilize tools before use    |

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## 4.10 Advanced Practice: Topworking

### What is Topworking?

Topworking is the process of **changing the variety of an existing plant** by grafting new scions onto mature branches.

#### Purpose:

- Upgrade fruit quality
- Change plant variety without replanting
- Improve productivity

#### Example Scenario:

- An old tree producing low-quality fruit
  - New scions with better traits are grafted onto it
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### Advantages

- Saves time compared to planting new trees
  - Utilizes established root systems
  - Faster production of improved crops
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## 4.11 Advanced Practice: Interstocks

### What is an Interstock?

An interstock is a **middle stem piece placed between the rootstock and scion**.

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### Why Use Interstocks?

## 1. Improve Compatibility

- Helps join plants that cannot graft directly

## 2. Control Growth

- Can influence plant size (e.g., dwarfing)

## 3. Modify Traits

- Can affect fruiting behavior and structure
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### Example Structure:

Scion (desired fruit)

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Interstock (compatibility or control)

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Rootstock (roots and vigor)

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## 4.12 Long-Term Care of Grafted Plants

### 1. Regular Monitoring

- Check graft union for damage or disease

### 2. Pruning

- Shape plant growth
- Maintain balance between rootstock and scion

### 3. Nutrient Management

- Ensure adequate fertilization
  - Support healthy growth
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## 4.13 Practical Demonstration Scenario

### Post-Graft Monitoring Timeline:

- **Week 1–2:** Maintain moisture and stability
  - **Week 3–4:** Look for bud swelling
  - **Month 2+:** Remove wraps if necessary, support new growth
  - **Long-term:** Prune and manage plant structure
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#### **4.14 Learning Outcomes**

By the end of this module, you should be able to:

- Manage grafts during the healing phase
- Identify successful and failed grafts
- Apply troubleshooting techniques
- Perform advanced practices like topworking
- Understand the role of interstocks
- Maintain long-term health of grafted plants